

AUTOMATED GRADING SYSTEM FOR ESSAY RESPONSES USING NLP

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ABSTRACT

This study develops an automated essay grading system using BERT sentence embeddings and cosine similarity scoring, achieving 0.87 Pearson correlation with expert human graders.

INTRODUCTION

Manual essay grading is time-consuming and subjective. Automated Essay Scoring (AES) systems can provide consistent, instant feedback. This study targets short-answer essay questions in STEM subjects at CvSU, where graders assess technical accuracy and conceptual understanding. The system compares student responses against model answers using semantic similarity.

METHODOLOGY

BERT sentence embeddings (bert-base-uncased, pooled CLS token) were used to encode both student responses and model answers. Cosine similarity between embeddings was mapped to a 0-10 score scale via linear regression trained on 500 expert-graded pairs. A rubric-alignment module penalizes off-topic responses using keyword overlap (Jaccard similarity). Dataset: 3,200 essay responses across 8 STEM courses.

RESULTS

The system achieved Pearson correlation $r=0.87$ with expert scores on the held-out test set. Mean absolute error was 0.71 points on a 10-point scale. Inter-rater agreement between system and human expert (Cohen's Kappa=0.79) was comparable to inter-human agreement (0.82). Processing time: 0.3 seconds per essay.

DISCUSSION

BERT-based semantic scoring outperforms TF-IDF cosine similarity ($r=0.71$) and keyword matching ($r=0.58$) for essay grading tasks. The main failure mode is factually correct responses phrased very differently from the model answer. Future work should fine-tune on domain-specific STEM corpora and incorporate factual consistency checking.